Activity 2.1 – Sequence Detector

1.Design Approach:

Designed Moore and Mealy sequence detectors using Verilog HDL to detect the binary sequence 1011. Implemented finite state machines to identify the required sequence and generate a detection output.

2.Verification Steps:

- Created separate testbench files.
- Applied reset to initialize the state machines.
- Provided the input sequence 1011.
- Tested sequence detection.
- Checked overlapping sequence detection.
- Simulated using ModelSim.

3.Results:

The testbench checks the detection outputs of both Moore and Mealy sequence detectors. PASS/FAIL messages are displayed.

4.Issues Faced:

State transitions must be defined correctly. Overlapping sequences require proper state management. Moore and Mealy output timing must be considered.

5.Conclusion:

Successfully designed and verified Moore and Mealy sequence detectors using Verilog HDL.